MODULE 18 · DOMAIN 4 - EVALUATION, TESTING AND OPTIMISATION

Eval Datasets and Test Frameworks

Building a dataset that represents real use, choosing graders, the test types and when each runs, and scoring that can be trusted across versions.

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An evaluation is a driving test with a fixed route. **Not it seemed fine on my street.**

Anthropic's vocabulary for evaluations

TERM	MEANING
Task	One test, with defined inputs and success criteria
Trial	One attempt at a task; run several, outputs vary
Grader	Logic that scores one aspect of a trial
Transcript	The full record: outputs, tool calls, reasoning
Outcome	The final state of the environment after the trial
Suite and harness	A set of tasks, and the system that runs them

ON THE EXAM

Grade the outcome where you can. A transcript shows how; the outcome shows whether.

What goes in the dataset

SLICE	WHAT IT COVERS
Representative	The real mix of requests, drawn from logs and records
Population coverage	Every product, channel, region and group served
Edge cases	Irrelevant, overlong, ambiguous or incomplete input
Adversarial	Injection, jailbreaks, policy probes
Balanced	Cases where a behaviour should, and should not, occur
Labels and rubric	Expert labels against a written rubric

ON THE EXAM

Anthropic: prioritise volume over quality. More cases with automated grading beat a few hand-graded ones.

Three kinds of grader

GRADER	STRENGTHS	WEAKNESSES	USE IT FOR
Code	Fast, cheap, objective, reproducible	Brittle, no nuance	Fields, formats, tool calls, exact values
Model	Flexible, scalable, catches nuance	Varies; must be calibrated	Meaning, tone, grounding, rubric checks
Human	Gold standard, expert judgement	Slow and expensive	Calibration, ambiguous and high-stakes cases

ON THE EXAM

Use the fastest grader that is reliable for each check. Test a model grader against humans before scaling it.

Test types, and when each runs

TEST	ANSWERS THE QUESTION	RUNS
Unit	Does this one prompt or tool behave?	While building
Integration	Does it work end to end, with real tools and retrieval?	Before release
Regression	Did a change break what worked?	On every change
Adversarial	Does it resist attack and misuse?	Before release, and on change
Smoke	Is the deployed system alive and sane?	Straight after deploy

ON THE EXAM

Name the question first. A change means regression. A deployment means smoke. An attack surface means adversarial.

Capability evals and regression evals

CAPABILITY EVALS

A hill to climb

- New or hard behaviour
- Start with a low pass rate
- Show whether changes help
- Graduate when they pass reliably

REGRESSION EVALS

A floor to protect

- Behaviour that already works
- Sit near a hundred percent
- Any drop blocks the release
- Grow as capabilities graduate

THE TIEBREAKER

When everything passes, the eval is saturated. Add harder tasks, or it can no longer show progress.

Scores you can compare across versions

- **A fixed, versioned set:** the same tasks for every candidate.
- **Record every run's versions:** model, prompt, tools, index, dataset.
- **Change one thing at a time,** or rerun the old set on the new system.
- **Several trials per task,** because outputs vary between runs.
- **Never change the set and the system together** and compare the scores.

> ON THE EXAM
>
> A score without versions attached is a rumour.

The evaluation sequence

01
Define
Metrics, thresholds and slices.

02
Curate, label
Real cases, expert rubric.

03
Run
Several trials, versions recorded.

04
Score
Code, model, human graders.

05
Review failures
Read the transcripts.

06
Gate release
Thresholds decide, not mood.

ON THE EXAM

Start small: Anthropic suggests twenty to fifty tasks drawn from real failures, then grow.

Evals are one layer, not the whole defence

- **Automated evals** before every release.
- **Production monitoring** for ground truth on real use.
- **A/B tests** for significant changes, with enough traffic.
- **Feedback and transcript review** for what nobody thought to test.
- **Periodic human studies** to recalibrate model graders.

> THE PICTURE
>
> Anthropic's Swiss cheese model: no single layer catches every issue, so stack them.

> Mereside's complaints triage evaluation set was built only from mortgage complaints. It scores well.
>
> The rollout now covers all complaints, including savings, branch service and complaints received by letter.

What should Corvane fix first?

A Add more mortgage complaints to strengthen the statistics

B Switch from model grading to human grading

C Have Claude generate synthetic savings complaints to fill the gap

D Rebuild the set from labelled historical complaints matching the real mix of types and channels, including rare categories

◆ PAUSE AND DECIDE

What should Corvane fix first?

A Add more mortgage complaints to strengthen the statistics

More of the one slice already covered.

B Switch from model grading to human grading

Better grading of the wrong population.

C Have Claude generate synthetic savings complaints to fill the gap

Useful later, but real labelled complaints exist.

D Rebuild the set from labelled historical complaints matching the real mix of types and channels, including rare categories

The set now represents who the system serves.

THE SIGNAL

An eval set must cover the population served. A high score on one slice proves nothing about the rest.

> Quillon's discharge summaries have structured fields, such as discharge date and medication list, and a narrative section.
>
> Corvane must choose how to grade about 800 evaluation cases on every release.

Which grading approach fits best?

A Clinicians grade every case by hand on every release

B Text overlap between each summary and a clinician-written reference

C One model grader for everything, with no rubric

D Code checks for the fields, and a model grader with a clinical rubric for the narrative, calibrated against clinicians on a sample

◆ PAUSE AND DECIDE

Which grading approach fits best?

A Clinicians grade every case by hand on every release

Gold standard, but too slow to run on every release.

B Text overlap between each summary and a clinician-written reference

Penalises correct rewording; misses wrong facts in familiar words.

C One model grader for everything, with no rubric

Uncalibrated, and wasted on exact fields.

D Code checks for the fields, and a model grader with a clinical rubric for the narrative, calibrated against clinicians on a sample

Each check gets the fastest grader that is reliable for it.

THE SIGNAL

Mixed outputs need mixed graders: code for exact parts, a calibrated model for meaning, humans to calibrate.

> Tallyhawk reworded several sections of its support agent's system prompt to improve tone.
>
> The change is ready for release next week.

Which test must pass before release?

A The regression suite on the fixed evaluation set, compared with the current release

B A smoke test of five questions after deployment

C An A/B test on live traffic before any offline testing

D A new capability evaluation for tone

◆ PAUSE AND DECIDE

Which test must pass before release?

A The regression suite on the fixed evaluation set, compared with the current release

A change was made: prove it broke nothing.

B A smoke test of five questions after deployment

After release, and far too shallow.

C An A/B test on live traffic before any offline testing

Exposes customers to an untested change.

D A new capability evaluation for tone

Useful too, but it cannot show nothing else broke.

THE SIGNAL

Any change to prompt, model, tools or retrieval: run the regression suite before release.

> Corvane uses a model grader to score the correctness of Mereside's agent-assist answers against policy. Its scores look high.
>
> Nobody has yet checked the grader itself.

What should Corvane do before relying on it?

A Use the same model and prompt that produced the answers as the grader

B Compare its verdicts with policy experts' on a sample, and refine the rubric until they agree

C Ask it for scores out of a hundred, for more precision

D Accept it, since high scores match the team's expectations

◆ PAUSE AND DECIDE

What should Corvane do before relying on it?

A. Use the same model and prompt that produced the answers as the grader
Marks its own homework, with the same blind spots.

B. Compare its verdicts with policy experts' on a sample, and refine the rubric until they agree
Calibration: proves the grader before trusting it.

C. Ask it for scores out of a hundred, for more precision
False precision. Specific verdicts grade better.

D. Accept it, since high scores match the team's expectations
Agreement with hopes is not validation.

THE SIGNAL

A model grader is a measuring instrument. Calibrate it against experts before you trust its readings.

> Quillon's portal evaluation checks that the assistant refuses clinical advice. All 200 test prompts ask for clinical advice, and it refuses every one.
>
> In the pilot, it also refuses legitimate questions, such as whether to fast before a booked blood test.

What is missing from the evaluation?

A More adversarial prompts asking for clinical advice

B A stricter refusal rule in the system prompt

C Cases near the boundary where the assistant should answer, scored alongside the refusal cases

D Human grading of the existing refusal cases

◆ PAUSE AND DECIDE

What is missing from the evaluation?

A More adversarial prompts asking for clinical advice

More of the one side already tested.

B A stricter refusal rule in the system prompt

Would make over-refusal worse.

C Cases near the boundary where the assistant should answer, scored alongside the refusal cases

A balanced set catches over-refusal as well as under-refusal.

D Human grading of the existing refusal cases

Better grading of an unbalanced set.

THE SIGNAL

Test both sides of every boundary: where the behaviour should happen, and where it should not.

> Tallyhawk's evaluation score fell from 91% to 86% between two runs.
>
> In between, the team added 120 new hard cases to the set and moved to a newer model version. Nobody can say which change caused the drop.

Which practice would have prevented the confusion?

A. Average the two runs and report the result

B. Keep a fixed, versioned set, record versions per run, and change one thing at a time

C. Remove the new hard cases so the score recovers

D. Report whichever run scored higher

◆ PAUSE AND DECIDE

Which practice would have prevented the confusion?

A. Average the two runs and report the result
Blends two different experiments into one meaningless number.

B. Keep a fixed, versioned set, record versions per run, and change one thing at a time
Every difference is attributable to one change.

C. Remove the new hard cases so the score recovers
Hides weakness to protect a number.

D. Report whichever run scored higher
Cherry-picking.

THE SIGNAL

Two things changed and the score moved: you cannot know why. Version the set, change one thing.

> Tallyhawk's billing agent evaluation checks the exact sequence of tool calls for each task.
>
> A new version solves many tasks correctly using a different, valid sequence, and fails the evaluation.

What should Corvane change?

A Force the agent to follow the old sequence of tool calls

B Lower the pass threshold for the evaluation

C Remove the affected tasks from the set

D Grade the final account state, and read transcripts to confirm the failures are real

◆ PAUSE AND DECIDE

What should Corvane change?

A Force the agent to follow the old sequence of tool calls

Fits the agent to a brittle grader.

B Lower the pass threshold for the evaluation

Lets real failures through too.

C Remove the affected tasks from the set

Loses coverage of real work.

D Grade the final account state, and read transcripts to confirm the failures are real

Grades what matters, the outcome, and checks the grader.

THE SIGNAL

Valid solutions failing the eval: grade outcomes, not paths, and read the transcripts.

> Ashgrove Insurance is building an evaluation for a new claims assistant before launch.
>
> Its team asks which steps must happen before the evaluation is run for the first time.

Which two steps come before the first run? Choose two.

A Define the metrics, thresholds and slices the set must cover

B Gate the release on the evaluation results

C Curate real cases and label them with experts against a written rubric

D Review the failures by reading transcripts

E Run an A/B test with live customers

◆ PAUSE AND DECIDE

Which two steps come before the first run? Choose two.

A Define the metrics, thresholds and slices the set must cover

Step one: decide what good means.

B Gate the release on the evaluation results

Needs results. It comes last.

C Curate real cases and label them with experts against a written rubric

Step two: the route the system will drive.

D Review the failures by reading transcripts

Needs a run to review.

E Run an A/B test with live customers

After launch, with enough traffic. Not before an offline eval.

THE SIGNAL

Define before you build, measure before you change: metrics and labelled data come first.

In five lines

- **The set:** representative, covering the population, with edge, adversarial and balanced slices.
- **Graders:** code for exact parts, calibrated model for meaning, humans to calibrate.
- **Test types match moments:** regression on change, smoke after deploy, adversarial for attack.
- **Comparable scores:** fixed versioned set, versions recorded, one change at a time.
- **Sequence:** define, curate, run, score, read transcripts, gate.

COMING UP

Next: A/B testing and iteration.